

Doubly Fair Dynamic Pricing

Complete Explanatory Notes

From First Principles to Implementation-Ready Understanding

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What these notes cover: A self-contained, mathematically rigorous, and intuitively grounded explanation of the paper — including all definitions, algorithms, theorems, and proof sketches — written so that anyone with basic probability and linear algebra can read, implement, and present this work confidently.

Contents

1	Overview and Motivation	4
1.1	The Big Picture in One Paragraph	4
1.2	Why This Problem Is Hard	4
1.3	Contributions at a Glance	5
2	Background: Classical Online Pricing	6
2.1	The Single-Product Pricing Protocol	6
2.2	The Revenue Function	7
2.3	What Makes This an “Online Learning” Problem	7
3	Fairness: Concepts and Definitions	8
3.1	Why Pricing Fairness Matters	8
3.2	Procedural Fairness (Equality of Application)	8
3.3	Substantive Fairness (Equality of Outcome)	9
3.4	Doubly Fair = Both at Once	9
4	Why Randomness is Necessary	10
4.1	The Impossibility of Deterministic Doubly-Fair Policies	10
4.2	The Motivating Example in Detail	11
4.2.1	Step 1: Best Deterministic Policy	11
4.2.2	Step 2: The Optimal Doubly-Fair Random Policy	11
4.2.3	Step 3: Verify Procedural Fairness	12
4.2.4	Step 4: Verify Substantive Fairness	12
4.2.5	Step 5: Compute Revenue and Compare	12

5	Formal Problem Setup	14
5.1	The Two-Group Online Pricing Process	14
5.2	Key Mathematical Objects	15
5.3	Four Fundamental Quantities for Group e	15
5.4	Definition: Expected Revenue	16
5.5	Definition: Procedural Unfairness	16
5.6	Definition: Substantive Unfairness	17
5.7	The Optimal Policy	17
5.8	Definitions: Regret and Cumulative Unfairness	18
5.9	Technical Assumptions	18
6	The Fairly Pricing Algorithm (FPA)	20
6.1	High-Level Architecture	20
6.2	Phase 0: Before Epochs — Estimating $F_{\min}$	20
6.3	Phase 1: Doubling Epochs	21
6.4	Phase 2: Good-and-Exploratory Policies	22
6.5	Phase 3: Probability Estimation	22
6.6	Phase 4: Policy Elimination (Algorithms 2 & 3)	23
6.6.1	Finding the Empirical Optimal Policy (Algorithm 2)	23
6.6.2	Eliminating Suboptimal Policies (Algorithm 3)	24
6.7	Complete Algorithm (Pseudocode)	24
7	Regret and Unfairness Analysis	25
7.1	Main Upper Bound Result	25
7.1.1	Proof Sketch (Main Ideas)	25
7.2	Regret Lower Bound	26
7.3	Substantive Unfairness Lower Bound	27
8	Discussion: Extensions and Open Problems	29
8.1	Technical Assumption Relaxation	29
8.2	Relaxed Substantive Fairness	29

8.3	Multi-Group Extension	30
8.4	Continuous Price Space	30
8.5	Trade-offs Between Fairness Types	30
9	Implementation Guide	31
9.1	Setting Up the Problem	31
9.2	Key Implementation Challenges	31
9.2.1	Linear Programming Subroutine	31
9.2.2	Grid Search for w	31
9.2.3	Tracking Active Price Sets	32
9.3	Complexity Summary	32
10	Mathematical Appendix: Detailed Proofs	33
10.1	Derivation of the Optimal Random Policy in Example 4.1	33
10.2	Hoeffding's Inequality (Used in Estimation Bounds)	34
10.3	Lipschitz Property of Substantive Unfairness	34
10.4	Why the Policy Set Representation is Tractable	35
11	Summary: The Complete Picture	36
11.1	What We Proved	36
11.2	Reading Roadmap for Implementers	37
11.3	Connections to Broader ML Themes	37

Chapter 1

Overview and Motivation

1.1 The Big Picture in One Paragraph

You run an online store selling identical products to two groups of customers — say women and men, or premium-loyalty users and regular users. You don’t know in advance what any individual is *willing to pay*. You want to maximize your total revenue. But you also want to be *fair*: the price you show women should not systematically differ from the price you show men (that is **procedural fairness**), and the price women actually *end up paying* should not systematically differ from what men pay (that is **substantive fairness**). This paper asks: *can we learn a pricing strategy, from customer interactions alone, that is simultaneously revenue-optimal and doubly fair?* The answer is **yes**, and the algorithm that does it is called **FPA** (Fairly Pricing Algorithm).

1.2 Why This Problem Is Hard

Three Sources of Difficulty

1. **Unknown demand.** We never observe a customer’s true willingness-to-pay y_t . We only see a binary signal: did they buy at our price or not? This is the “censored feedback” problem.
2. **Two fairness constraints simultaneously.** Prior work showed that you cannot be both procedurally and substantively fair with deterministic prices (unless you charge everyone the same price, sacrificing revenue). The fix – random pricing policies – introduces non-convexity into the optimization.
3. **Non-convex objective.** The substantive fairness constraint involves a ratio of

expected values (conditional expectation), making the feasible set non-convex and hard to optimize over.

1.3 Contributions at a Glance

1. **Algorithm FPA** achieves (with probability $\geq 1 - \varepsilon$):

$$\text{Regret} = \tilde{O}\left(\sqrt{T} d^{3/2}\right), \quad \text{Procedural Unfairness} = 0, \quad \text{Substantive Unfairness} = \tilde{O}\left(\sqrt{T} d^{3/2}\right).$$

2. **Regret lower bound** $\Omega(\sqrt{dT})$: FPA is information-theoretically optimal in T up to $\log \log T$ factors.
3. **Unfairness lower bound** $\Omega(\sqrt{T})$: *any* regret-optimal algorithm must suffer at least this much substantive unfairness. FPA matches this too.
4. **First** dynamic pricing algorithm with *two* simultaneous fairness constraints and provable optimality.

Chapter 2

Background: Classical Online Pricing

2.1 The Single-Product Pricing Protocol

Before fairness enters the picture, let us understand the classical online pricing problem that this paper builds upon.

Online Pricing Protocol (Classical)

For each round $t = 1, 2, \dots, T$:

1. A customer arrives and privately forms a valuation y_t (drawn i.i.d. from an unknown distribution D). We never observe y_t directly.
2. The seller (us) proposes a price v_t *without* knowing y_t .
3. The customer buys if and only if $v_t \leq y_t$, i.e., the purchase decision is:

$$\mathbf{1}_t = \mathbf{1}(v_t \leq y_t) \in \{0, 1\}.$$

4. The seller receives revenue $r_t = v_t \cdot \mathbf{1}_t$.

Think of it like this: you're guessing someone's "secret number" (their willingness to pay). You can guess any value. They'll only tell you "yes I'd pay that" or "no that's too much." Over time, you narrow down the range, but you can never be sure you've found the exact optimum.

2.2 The Revenue Function

For a fixed price v and demand distribution D , define the **acceptance probability** (also called the complementary CDF of D):

$$F(v) = \mathbb{P}_D[y \geq v].$$

Note that F is non-increasing (higher price $\Rightarrow$ fewer buyers). The **expected revenue** at price v is:

$$R(v) = v \cdot F(v).$$

This is a classic inverted-U shaped function in many settings: too low a price sells to everyone but earns little; too high a price earns a lot per sale but almost no one buys. The goal is to find the optimal price $v^* = \arg \max_v R(v)$ without knowing F in advance.

2.3 What Makes This an “Online Learning” Problem

Because D is unknown, we cannot compute v^* upfront. Instead, we learn it from the binary feedback $\{\mathbf{1}_t\}$. This is an **online bandit** problem: we receive partial feedback (not a full gradient of the revenue function) and must simultaneously explore (try different prices to learn F) and exploit (propose good prices to earn revenue).

Classical regret bounds (single-group, no fairness):

- Noise-free valuations: $\Theta(\log \log T)$ regret (Kleinberg & Leighton, 2003).
- Stochastic/adversarial: $\Theta(\sqrt{T})$ or $\Theta(T^{2/3})$ regret.
- With a discrete price set of size d : bandit-style $\Omega(\sqrt{dT})$ lower bound.

This paper works with a **finite price set** $V = \{v_1, \dots, v_d\}$, so the relevant baseline is the $\Omega(\sqrt{dT})$ lower bound.

Chapter 3

Fairness: Concepts and Definitions

3.1 Why Pricing Fairness Matters

When a seller uses different pricing strategies for different demographic groups, two distinct harms can arise:

Harm 1 (Procedural): A woman browses an online store and sees prices that are, on average, higher than what a man would see for the identical product. The algorithm *offers* different prices based on group membership.

Harm 2 (Substantive): Even if prices are offered identically, women who *actually buy* the product pay more on average than men who buy it, because the pricing strategy is tuned to extract more surplus from one group's buying patterns.

3.2 Procedural Fairness (Equality of Application)

Procedural fairness (also called “price fairness” in prior work) says: the *distribution* of prices proposed to each group should be equal in expectation.

Procedural Fairness – Formal Definition

A pricing policy $\pi = (\pi^1, \pi^2)$ is **procedurally fair** if:

$$\mathbb{E}_{v \sim \pi^1}[v] = \mathbb{E}_{v \sim \pi^2}[v],$$

i.e., the expected proposed price is the same across groups.

This is the “justice is blind” notion: treat everyone by the same standard. It does *not* mean proposing *exactly* the same price every time — it means the *average* proposed price is equal across groups.

3.3 Substantive Fairness (Equality of Outcome)

Substantive fairness goes further: it requires that customers who *actually purchase* the product pay the same expected price, regardless of their group. This is about actual economic outcomes, not just what prices were offered.

Substantive Fairness – Formal Definition

A pricing policy $\pi = (\pi^1, \pi^2)$ is **substantively fair** with respect to demand distributions F^1, F^2 if:

$$\mathbb{E}[v^1 \mid v^1 \text{ accepted}] = \mathbb{E}[v^2 \mid v^2 \text{ accepted}],$$

i.e., the expected price *given that it was accepted* is equal across groups.

Imagine a store that randomly gives 50% discounts. Both Group A and Group B are equally likely to see either price (procedural fairness). But if Group A tends to buy more at the regular price while Group B mostly buys only at the discount, then Group A ends up paying more on average (substantive unfairness). Procedural fairness does not guarantee substantive fairness, and vice versa.

3.4 Doubly Fair = Both at Once

A policy is **doubly fair** if and only if it is simultaneously procedurally *and* substantively fair. This paper is the first to design a learning algorithm that achieves near-optimal revenue while maintaining double fairness.

Chapter 4

Why Randomness is Necessary

4.1 The Impossibility of Deterministic Doubly-Fair Policies

A key insight of this paper — proved before any algorithm is proposed — is that if we restrict ourselves to *deterministic* prices (proposing a fixed price to each group), then the only doubly-fair strategy is to charge the *same* fixed price to both groups. Anything more sophisticated requires randomness.

Key Structural Result (Informal)

A deterministic, doubly-fair pricing policy must propose the *same* price to both groups. Any improvement over this trivial strategy requires randomizing prices.

Why? With deterministic prices v^1 for Group 1 and v^2 for Group 2:

- Procedural fairness requires: $v^1 = v^2$ (since expected value of a point mass is the point itself).
- So the only procedurally fair deterministic policies are “propose the same price to everyone.”
- When $v^1 = v^2 = v$, the expected accepted price in both groups is just v (since acceptance is binary at a fixed price), so substantive fairness is automatically satisfied.

Therefore, the best we can do with deterministic prices is optimize the single price v over the weighted revenue function. But this can be *strictly worse* than the optimum over random policies.

4.2 The Motivating Example in Detail

This example drives the entire paper.

Example 4.1 (The Paper's Running Example)

Two customer groups: G_1 (30% of customers) and G_2 (70% of customers). Three possible prices: $\{0.625, 0.70, 1.00\}$. Acceptance rates:

	\$0.625	\$0.70	\$1.00
G_1 (30%)	3/5	1/2	1/2
G_2 (70%)	4/5	4/5	1/2

4.2.1 Step 1: Best Deterministic Policy

With a single fixed price for both groups, the expected revenue is the weighted sum of each group's expected revenue:

$$R(v) = 0.3 \cdot v \cdot F^1(v) + 0.7 \cdot v \cdot F^2(v).$$

Computing for each price:

$$R(0.625) = 0.3 \cdot 0.625 \cdot \frac{3}{5} + 0.7 \cdot 0.625 \cdot \frac{4}{5} = 0.3 \cdot 0.375 + 0.7 \cdot 0.5 = 0.1125 + 0.35 = 0.4625,$$

$$R(0.70) = 0.3 \cdot 0.70 \cdot \frac{1}{2} + 0.7 \cdot 0.70 \cdot \frac{4}{5} = 0.3 \cdot 0.35 + 0.7 \cdot 0.56 = 0.105 + 0.392 = 0.497,$$

$$R(1.00) = 0.3 \cdot 1.00 \cdot \frac{1}{2} + 0.7 \cdot 1.00 \cdot \frac{1}{2} = 0.3 \cdot 0.5 + 0.7 \cdot 0.5 = 0.15 + 0.35 = 0.50.$$

The best deterministic doubly-fair policy is $v = \$1.00$ with expected revenue **\$0.50**.

4.2.2 Step 2: The Optimal Doubly-Fair Random Policy

Now suppose we allow randomization. For each group, we propose a price distribution (a mixture of prices). The paper shows that the following policy satisfies both fairness constraints and achieves *strictly higher* revenue:

Optimal Doubly-Fair Random Policy for Example 4.1

Group 1 (G_1): propose \$0.625 with probability $\frac{20}{29}$ and \$1.00 with probability $\frac{9}{29}$.

Group 2 (G_2): propose \$0.70 with probability $\frac{25}{29}$ and \$1.00 with probability $\frac{4}{29}$.

4.2.3 Step 3: Verify Procedural Fairness

Expected proposed price for each group:

$$\begin{aligned}\mathbb{E}_{\pi^1}[v] &= \frac{20}{29} \cdot 0.625 + \frac{9}{29} \cdot 1.00 = \frac{12.5}{29} + \frac{9}{29} = \frac{21.5}{29} \approx 0.7414, \\ \mathbb{E}_{\pi^2}[v] &= \frac{25}{29} \cdot 0.70 + \frac{4}{29} \cdot 1.00 = \frac{17.5}{29} + \frac{4}{29} = \frac{21.5}{29} \approx 0.7414.\end{aligned}$$

Both equal $\frac{21.5}{29} = \frac{43}{58}$. **Procedural fairness: satisfied. ✓**

4.2.4 Step 4: Verify Substantive Fairness

The expected accepted price in group e is $\frac{\text{expected revenue}}{\text{expected acceptance rate}}$.

For Group 1:

$$\begin{aligned}\mathbb{E}[\text{revenue}, G_1] &= \frac{20}{29} \cdot 0.625 \cdot \frac{3}{5} + \frac{9}{29} \cdot 1.00 \cdot \frac{1}{2} \\ &= \frac{20}{29} \cdot 0.375 + \frac{9}{29} \cdot 0.5 = \frac{7.5}{29} + \frac{4.5}{29} = \frac{12}{29}, \\ \mathbb{E}[\text{acceptance}, G_1] &= \frac{20}{29} \cdot \frac{3}{5} + \frac{9}{29} \cdot \frac{1}{2} = \frac{12}{29} + \frac{4.5}{29} = \frac{16.5}{29} = \frac{33}{58}, \\ \mathbb{E}[v \mid \text{accepted in } G_1] &= \frac{12/29}{33/58} = \frac{12}{29} \cdot \frac{58}{33} = \frac{24}{33} = \frac{8}{11}.\end{aligned}$$

For Group 2:

$$\begin{aligned}\mathbb{E}[\text{revenue}, G_2] &= \frac{25}{29} \cdot 0.70 \cdot \frac{4}{5} + \frac{4}{29} \cdot 1.00 \cdot \frac{1}{2} \\ &= \frac{25}{29} \cdot 0.56 + \frac{4}{29} \cdot 0.5 = \frac{14}{29} + \frac{2}{29} = \frac{16}{29}, \\ \mathbb{E}[\text{acceptance}, G_2] &= \frac{25}{29} \cdot \frac{4}{5} + \frac{4}{29} \cdot \frac{1}{2} = \frac{20}{29} + \frac{2}{29} = \frac{22}{29}, \\ \mathbb{E}[v \mid \text{accepted in } G_2] &= \frac{16/29}{22/29} = \frac{16}{22} = \frac{8}{11}.\end{aligned}$$

Both equal $\frac{8}{11}$. **Substantive fairness: satisfied. ✓**

4.2.5 Step 5: Compute Revenue and Compare

$$R(\pi) = 0.3 \cdot \frac{12}{29} + 0.7 \cdot \frac{16}{29} = \frac{3.6}{29} + \frac{11.2}{29} = \frac{14.8}{29} = \frac{74}{145} \approx \$0.5103.$$

This exceeds the best deterministic policy revenue of \$0.50. **The random policy generates strictly more revenue while being doubly fair.**

Randomization allows us to “split the difference” between what’s optimal for each group separately, in a way that the averages (proposed and accepted prices) line up across groups. This wouldn’t be possible with fixed prices. The algebra looks magical, but the intuition is: different groups respond to prices differently, and by mixing prices in carefully chosen proportions, we can equate the conditional averages while extracting more revenue than any uniform price would.

Chapter 5

Formal Problem Setup

5.1 The Two-Group Online Pricing Process

Problem Description

- **Time horizon:** T rounds (known in advance).
- **Two groups:** G_1 and G_2 . Group G_1 comprises a fraction $q \in (0, 1)$ of all customers; G_2 comprises $1 - q$.
- **Price set:** $V = \{v_1, v_2, \dots, v_d\}$ with $0 < v_1 < v_2 < \dots < v_d \leq 1$. (Discrete, finite, and known.)
- **Pricing policy:** A tuple $\pi = (\pi^1, \pi^2)$ where $\pi^e \in \Delta^d$ (the probability simplex over d prices) for $e \in \{1, 2\}$. This means at each round, we choose a price for group G_e by sampling from π^e .

At each round t :

1. A customer from group G_{e_t} arrives (where e_t is observable/known to us).
2. We sample a price $v_t^{e_t} \sim \pi^{e_t}$ from V .
3. The customer's valuation $y_t^{e_t} \sim D^{e_t}$ is drawn independently from an unknown distribution D^e specific to group e .
4. We observe the binary feedback $\mathbf{1}_t^{e_t} = \mathbf{1}(v_t^{e_t} \leq y_t^{e_t})$.
5. We receive revenue $r_t^{e_t} = \mathbf{1}_t^{e_t} \cdot v_t^{e_t}$.

5.2 Key Mathematical Objects

Let us carefully define the notation that appears throughout the paper.

Notation Setup

$$\begin{aligned}\mathbf{v} &:= [v_1, v_2, \dots, v_d]^\top \in \mathbb{R}^d \quad (\text{column vector of prices}), \\ [d] &:= \{1, 2, \dots, d\} \quad (\text{index set}), \\ \mathbf{1} &:= [1, 1, \dots, 1]^\top \in \mathbb{R}^d \quad (\text{all-ones vector}).\end{aligned}$$

For group $e \in \{1, 2\}$ and price index $i \in [d]$:

$$F^e(i) := \mathbb{P}_{D^e}[y_t^e \geq v_i] \quad (\text{probability that price } v_i \text{ is accepted in group } G_e).$$

Since $v_1 < v_2 < \dots < v_d$, we have $F^e(1) \geq F^e(2) \geq \dots \geq F^e(d)$. Define the diagonal matrix:

$$\mathbf{F}_e := \text{diag}(F^e(1), F^e(2), \dots, F^e(d)) \in \mathbb{R}^{d \times d}.$$

$F^e(i)$ is the fraction of Group e customers who would agree to pay price v_i or more. Think of it as the complementary CDF of Group e 's valuation distribution, evaluated at each discrete price point. The matrix $\mathbf{F}_e$ collects all these values on a diagonal so that matrix-vector products give weighted averages naturally.

5.3 Four Fundamental Quantities for Group e

Using the notation above, for a customer from G_e under policy π :

Four Key Quantities

$$\text{Expected proposed price: } \mathbf{v}^\top \pi^e \tag{5.1}$$

$$\text{Expected revenue: } \mathbf{v}^\top \mathbf{F}_e \pi^e \tag{5.2}$$

$$\text{Expected acceptance rate: } \mathbf{1}^\top \mathbf{F}_e \pi^e \tag{5.3}$$

$$\text{Expected accepted price: } \frac{\mathbf{v}^\top \mathbf{F}_e \pi^e}{\mathbf{1}^\top \mathbf{F}_e \pi^e} \tag{5.4}$$

Derivation of (5.1): The price proposed is v_i with probability π_i^e , so the expected

proposed price is $\sum_i v_i \pi_i^e = \mathbf{v}^\top \pi^e$.

Derivation of (5.2): The revenue from price v_i is $v_i \cdot F^e(i)$ (price times probability of acceptance). Weighting by the policy: $\sum_i v_i F^e(i) \pi_i^e = \mathbf{v}^\top \mathbf{F}_e \pi^e$.

Derivation of (5.3): The probability a customer accepts is $\sum_i F^e(i) \pi_i^e = \mathbf{1}^\top \mathbf{F}_e \pi^e$.

Derivation of (5.4): The expected accepted price is the expected revenue *conditioned on* acceptance. By Bayes' rule:

$$\mathbb{E}[v \mid \text{accepted}] = \frac{\mathbb{E}[v \cdot \mathbf{1}(\text{accepted})]}{\mathbb{P}[\text{accepted}]} = \frac{\mathbf{v}^\top \mathbf{F}_e \pi^e}{\mathbf{1}^\top \mathbf{F}_e \pi^e}.$$

5.4 Definition: Expected Revenue

Definition 5.1 – Expected Revenue

For any pricing policy $\pi = (\pi^1, \pi^2)$, its **expected revenue** is:

$$R(\pi; F^1, F^2) := q \cdot \mathbf{v}^\top \mathbf{F}_1 \pi^1 + (1 - q) \cdot \mathbf{v}^\top \mathbf{F}_2 \pi^2 \quad (5.5)$$

This is the population-weighted average of each group's expected revenue.

5.5 Definition: Procedural Unfairness

Definition 5.2 – Procedural Unfairness

The **procedural unfairness** of policy π is:

$$U(\pi) := |\mathbf{v}^\top \pi^1 - \mathbf{v}^\top \pi^2| = |\mathbf{v}^\top (\pi^1 - \pi^2)| \quad (5.6)$$

A policy is procedurally fair iff $U(\pi) = 0$.

Notice that $U(\pi)$ depends only on the policy π itself — not on the unknown demand matrices F^1, F^2 . Since we choose π , we have *full control* over procedural unfairness. We can enforce $U(\pi) = 0$ exactly at every round, without needing to learn anything. This is why FPA achieves *zero* procedural unfairness (not just small).

Define the set of all procedurally fair policies:

$$\Pi := \{\pi = (\pi^1, \pi^2) : U(\pi) = 0\} = \{\pi : \mathbf{v}^\top \pi^1 = \mathbf{v}^\top \pi^2\}.$$

5.6 Definition: Substantive Unfairness

Definition 5.3 – Substantive Unfairness

The **substantive unfairness** of policy $\pi \in \Pi$ is:

$$S(\pi; F^1, F^2) := \left| \frac{\mathbf{v}^\top \mathbf{F}_1 \pi^1}{\mathbf{1}^\top \mathbf{F}_1 \pi^1} - \frac{\mathbf{v}^\top \mathbf{F}_2 \pi^2}{\mathbf{1}^\top \mathbf{F}_2 \pi^2} \right| \quad (5.7)$$

A policy is substantively fair iff $S(\pi; F^1, F^2) = 0$.

Unlike procedural unfairness, $S(\pi; F^1, F^2)$ depends on the unknown matrices F^1, F^2 through a *ratio* of expectations. This has two consequences:

1. We cannot enforce $S = 0$ exactly (since we don't know F^1, F^2).
2. The constraint $S = 0$ is **non-convex** (ratio of linear functions), making the optimization very hard.

This is why FPA can only achieve *small but nonzero* substantive unfairness.

5.7 The Optimal Policy

Definition 5.4 – Optimal Doubly-Fair Policy

$$\begin{aligned} \pi^* = \arg \max_{\pi=(\pi^1, \pi^2) \in \Pi} & R(\pi; F^1, F^2) \\ \text{subject to: } & U(\pi) = 0, \quad S(\pi; F^1, F^2) = 0. \end{aligned} \quad (5.8)$$

Feasibility is trivial: The policy $\pi^1 = \pi^2 = e_i$ (propose the same fixed price v_i to everyone) is always doubly fair (trivially satisfies both constraints). So π^* always exists.

Finding π^* is hard: Even if F^1, F^2 were known, the problem is non-convex and non-quasi-convex due to the ratio in the substantive fairness constraint. We'll discuss the computational solution in the algorithm section.

5.8 Definitions: Regret and Cumulative Unfairness

Definition 5.5 – Regret

For an algorithm $\mathcal{A}$ that proposes policy π_t at round t , the **cumulative regret** is:

$$\text{Reg}_T(\mathcal{A}) := \sum_{t=1}^T [R(\pi^*; F^1, F^2) - R(\pi_t; F^1, F^2)] . \quad (5.9)$$

Definition 5.6 – Cumulative Substantive Unfairness

The **cumulative substantive unfairness** of algorithm $\mathcal{A}$ is:

$$S_T(\mathcal{A}) := \sum_{t=1}^T S(\pi_t; F^1, F^2). \quad (5.10)$$

Note that $R(\pi_t; F^1, F^2) > R(\pi^*; F^1, F^2)$ is possible if π_t violates the fairness constraints (since π^* is optimal only within the fair policy set Π). So per-round regret can be negative. Only *cumulative* regret is guaranteed to be non-negative for reasonable algorithms.

5.9 Technical Assumptions

Assumption 5.1 – Least Probability of Acceptance

There exists a constant $F_{\min} > 0$ such that:

$$F^e(d) \geq F_{\min} \quad \text{for } e = 1, 2.$$

Why needed? The substantive unfairness formula involves dividing by $\mathbf{1}^\top \mathbf{F}_e \pi^e$ (the expected acceptance rate). If this could be zero, the formula is undefined. Assuming $F_{\min} > 0$ ensures the highest price v_d is still accepted with positive probability, bounding the denominators away from zero. It also implies that S is Lipschitz (small changes in the policy cause bounded changes in unfairness).

Assumption 5.2 – Number of Prices

The number of prices d is independent of T , and:

$$d = O(T^{1/3}).$$

Why needed? The lower bound proof requires this to apply a bandit-style $\Omega(\sqrt{dT})$ argument. Since we care about how fairness affects the time dependence (not the d dependence), this separates the two.

Chapter 6

The Fairly Pricing Algorithm (FPA)

6.1 High-Level Architecture

FPA is a **policy-elimination algorithm** based on the “doubling epoch” framework. The key ideas are:

1. Maintain a set Π_k of “candidate” policies that might be optimal.
2. In each epoch, use the current candidates to *explore* (estimate F^1, F^2 better) and *exploit* (propose the best-estimated policy).
3. At the end of each epoch, *eliminate* policies that are clearly suboptimal or unfair.
4. Double the epoch length each time, so later epochs have better estimates and thus better policies.

In a standard multi-armed bandit, one could try each arm independently. Here, the “arms” are entire pricing policies $\pi \in \Pi$ — an infinite set (all distributions over d prices). We can’t try each one. Instead, FPA identifies a small “representative” set of *good-and-exploratory* policies that let us estimate the quality of all other policies indirectly.

6.2 Phase 0: Before Epochs — Estimating $F_{\min}$

Purpose: Estimate a lower bound on $F_{\min}$, which is needed for the algorithm’s oracle subroutines.

Method: Propose the highest price v_d for $\tau_0 = 2 \log T \cdot \log(16/\varepsilon)$ rounds. For each group e , count:

- $M_{0,e}$: number of times a customer from group e appeared.
- $N_{0,e}$: number of times they accepted price v_d .

Output the estimate:

$$\hat{F}_{\min} = \min \left\{ \frac{N_{0,1}}{2M_{0,1}}, \frac{N_{0,2}}{2M_{0,2}} \right\}.$$

Why halve the empirical rate? The factor of $1/2$ makes this a conservative (lower) estimate. The concentration inequality guarantees that with high probability, $\hat{F}_{\min} \leq F_{\min}$ (true value).

Why propose v_d specifically? Since prices are ordered and acceptance rates decrease with price, $F^e(d)$ is the *minimum* acceptance rate. Proposing v_d gives us a direct estimate of the hardest-to-accept price.

6.3 Phase 1: Doubling Epochs

Structure: After the before-epochs phase, time is divided into epochs $k = 1, 2, 3, \dots$. Each epoch k has length:

$$\tau_k = O(\sqrt{T} \cdot 2^k).$$

Thus: $\tau_1 < \tau_2 < \tau_3 < \dots$ (doubling each time).

Why doubling? Early epochs are short: we have little data, so estimates are rough, and good policies haven't been identified yet. Later epochs are long: estimates are accurate, the policy set is narrow, and the algorithm is close to optimal. The doubling ensures that the total time across all epochs is $O(T)$, and the geometric growth means most time is spent in later (better) epochs.

Number of epochs: Since $\sum_{k=1}^K \tau_k = T$, we have $K = O(\log T)$ epochs.

Per-epoch parameters (set at the start of epoch k):

$$\begin{aligned} \delta_{k,r} &= O \left(\sqrt{\frac{d^3 \log d \log(1/\varepsilon)}{\tau_k}} \right) && \text{(reward uncertainty),} \\ \delta_{k,s} &= O \left(\sqrt{\frac{d^3 \log d \log(1/\varepsilon)}{\tau_k}} \right) && \text{(substantive unfairness uncertainty).} \end{aligned}$$

Both shrink as τ_k grows, reflecting that longer epochs yield better estimates.

6.4 Phase 2: Good-and-Exploratory Policies

Purpose: Identify a small set A_k of policies that (i) are potentially near-optimal (in the current candidate set Π_k) and (ii) explore all relevant prices.

Core idea: For each price v_i and each group e , find the policy in Π_k that *maximizes* the probability of proposing v_i to group G_e :

$$\tilde{\pi}_{k,i,e} = \arg \max_{\pi \in \Pi_k} \pi^e(i).$$

If the maximum probability is at least $\frac{1}{\sqrt{T}}$, include $\tilde{\pi}_{k,i,e}$ in A_k . Otherwise, price v_i is so unlikely under any good policy that estimating $F^e(i)$ isn't worth the exploration cost; remove i from the active index set I_k^e .

The policy $\tilde{\pi}_{k,i,e}$ is the “most exploratory” policy for price v_i in group e , within the set of remaining candidates. Running it for enough rounds gives a reliable estimate of $F^e(i)$. The threshold $\frac{1}{\sqrt{T}}$ ensures that even if we run this policy for only $\frac{\tau_k}{|A_k|}$ rounds, we still see price v_i proposed to group e a reasonable number of times.

Size of A_k : At most $2d$ policies (two groups times up to d prices). So $|A_k| \leq 2d$.

6.5 Phase 3: Probability Estimation

Purpose: Use the policies in A_k to estimate $F^1(i)$ and $F^2(i)$ for all active price indices.

Method: Run each policy $\pi \in A_k$ for a batch of $\tau_k/|A_k|$ rounds. During these rounds, count:

- $M_{k,e}(i)$: times price v_i was proposed to group G_e .
- $N_{k,e}(i)$: times price v_i was accepted in group G_e .

The empirical estimate of $F^e(i)$ is:

$$\bar{F}_{k,e}(i) = \max \left\{ \frac{N_{k,e}(i)}{M_{k,e}(i)}, \hat{F}_{\min} \right\} \quad \text{for } i \in I_k^e,$$

and $\bar{F}_{k,e}(i) = \hat{F}_{\min}$ for $i \notin I_k^e$.

Why clamp at $\hat{F}_{\min}$? The assumption guarantees $F^e(i) \geq F_{\min}$, so clamping ensures our estimate is also bounded away from zero (avoiding division-by-zero issues in the substantive

unfairness formula).

Concentration bound: By Hoeffding's inequality, since each good-and-exploratory policy proposes v_i to group e with probability $\tilde{\pi}_{k,i,e}^e(i) \geq \frac{1}{\sqrt{T}}$, and the batch runs for $\tau_k/|A_k|$ rounds, the number of observations $M_{k,e}(i)$ is at least $\tau_k/(|A_k| \cdot \sqrt{T}) = \Omega(\sqrt{\tau_k/T} \cdot \tau_k/d)$ with high probability. This gives the estimation error $|\bar{F}_{k,e}(i) - F^e(i)| = O(\delta_{k,s})$.

6.6 Phase 4: Policy Elimination (Algorithms 2 & 3)

6.6.1 Finding the Empirical Optimal Policy (Algorithm 2)

Given the estimated matrices $\hat{\mathbf{F}}_{k,e}$, find the policy that maximizes estimated revenue under an estimated substantive fairness constraint:

$$\begin{aligned} \hat{\pi}_{k,*} &= \arg \max_{\pi \in \Pi_k} R(\pi, \hat{\mathbf{F}}_{k,1}, \hat{\mathbf{F}}_{k,2}) \\ \text{subject to: } & S(\pi, \hat{\mathbf{F}}_{k,1}, \hat{\mathbf{F}}_{k,2}) \leq \delta_{k,s}. \end{aligned} \quad (6.1)$$

Computational challenge: Problem (6.1) is non-convex because S is a ratio.

The linearization trick: Introduce a scalar auxiliary variable $w = \mathbb{E}[v \mid \text{accepted in } G_1]$ (the target accepted price). For a fixed value of w , the constraint $S \leq \delta_{k,s}$ becomes:

$$\begin{aligned} \mathbf{v}^\top \mathbf{F}_1 \pi^1 &= w \cdot \mathbf{1}^\top \mathbf{F}_1 \pi^1, \\ (w - \delta_{k,s}) \cdot \mathbf{1}^\top \mathbf{F}_2 \pi^2 &\leq \mathbf{v}^\top \mathbf{F}_2 \pi^2 \leq (w + \delta_{k,s}) \cdot \mathbf{1}^\top \mathbf{F}_2 \pi^2. \end{aligned}$$

These are *linear* constraints in π^1 and π^2 (for fixed w)! So for each fixed w , we solve a linear program. Then we grid-search over $w \in [0, 1/\hat{F}_{\min}]$ with step size $\delta_{k,s}/2$, and take the best solution.

The key insight: although S is non-convex in (π^1, π^2, w) jointly, fixing w makes everything linear. We convert an infinite-dimensional non-convex problem into a finite number ($O(1/\delta_{k,s})$ many) of linear programs, each solvable in polynomial time. The Lipschitzness of S ensures the grid search loses at most $O(\delta_{k,s})$ in accuracy.

6.6.2 Eliminating Suboptimal Policies (Algorithm 3)

Update the candidate set to:

$$\Pi_{k+1} = \left\{ \pi \in \Pi_k : S(\pi, \hat{\mathbf{F}}_{k,1}, \hat{\mathbf{F}}_{k,2}) \leq \delta_{k,s}, \text{ and } R(\pi, \hat{\mathbf{F}}_{k,1}, \hat{\mathbf{F}}_{k,2}) \geq R(\hat{\pi}_{k,*}, \hat{\mathbf{F}}_{k,1}, \hat{\mathbf{F}}_{k,2}) - \delta_{k,r} - L \cdot \delta_{k,s} \right\}. \quad (6.2)$$

Two elimination criteria:

1. **Large estimated unfairness:** $S(\pi, \hat{F}) > \delta_{k,s}$. We're confident these policies are substantively unfair.
2. **Low estimated revenue:** $R(\pi, \hat{F}) < R(\hat{\pi}_{k,*}, \hat{F}) - \delta_{k,r} - L \cdot \delta_{k,s}$. These policies are suboptimal even accounting for estimation error.

Why the two subtractors on revenue?

- $\delta_{k,r}$ accounts for estimation error: the true revenue of a policy might differ from our estimate by up to $\delta_{k,r}$.
- $L \cdot \delta_{k,s}$ accounts for the relaxation: by allowing $S \leq \delta_{k,s}$ instead of $S = 0$, the optimal revenue could increase by at most $L \cdot \delta_{k,s}$ (by the Lipschitz property of the problem). We subtract this to ensure the true optimal π^* is never eliminated.

Conservation guarantee: The optimal policy π^* always remains in Π_k for all k . (This is the core induction argument in the proof.)

6.7 Complete Algorithm (Pseudocode)

Algorithm 1 Fairly Pricing Algorithm (FPA)

Require: Time horizon T , price set V , error probability ε , constant L , proportion q

- 1: **Before Epochs:** Propose highest price v_d for $\tau_0 = 2 \log T \cdot \log(16/\varepsilon)$ rounds; compute $\hat{F}_{\min}$
 - 2: Initialize: $\Pi_1 = \Pi$ (all procedurally fair policies), $I_0^1 = I_0^2 = [d]$
 - 3: **for** epoch $k = 1, 2, 3, \dots$ **do**
 - 4: Set parameters: $\tau_k, \delta_{k,r}, \delta_{k,s}$
 - 5: **Step 1 (Explore):** Select good-and-exploratory set A_k from Π_k
 - 6: **Step 2 (Estimate):** Run each $\pi \in A_k$ for $\tau_k/|A_k|$ rounds; compute $\bar{F}_{k,e}(i)$ for all e, i
 - 7: **Step 3 (Find Optimum):** Solve (6.1) using Algorithm 2; get $\hat{\pi}_{k,*}$
 - 8: **Step 4 (Eliminate):** Compute Π_{k+1} via (6.2) using Algorithm 3
 - 9: **end for**
-

Chapter 7

Regret and Unfairness Analysis

7.1 Main Upper Bound Result

Theorem 7.1 (Regret and Unfairness of FPA)

With probability at least $1 - \varepsilon$, FPA achieves:

$$\begin{aligned}\text{Reg}_T(\text{FPA}) &= O\left(\sqrt{T} d^{3/2} \cdot \frac{\log d \cdot \log T}{\varepsilon}\right), \\ U(\pi_t) &= 0 \quad \text{for all } t, \\ S_T(\text{FPA}) &= O\left(\sqrt{T} d^{3/2} \cdot \frac{\log d \cdot \log T}{\varepsilon}\right).\end{aligned}$$

What this means: Both regret and cumulative substantive unfairness grow as $\tilde{O}(\sqrt{T})$, while procedural unfairness is maintained at exactly zero throughout. The $\sqrt{T}$ growth means the *average* per-round regret and unfairness both shrink as $1/\sqrt{T}$, vanishing as $T \rightarrow \infty$. These are order-optimal results.

7.1.1 Proof Sketch (Main Ideas)

The proof proceeds by induction over epochs, establishing the following claim for all k :

Induction Claim for Epoch k

- (a) $\pi^* \in \Pi_k$ (optimal policy never eliminated).
- (b) For all π remaining in Π_k : $R(\pi^*, F^1, F^2) - R(\pi, F^1, F^2) \leq C (\delta_{k,r} + L \delta_{k,s})$.
- (c) For all $\pi \in \Pi_k$: $S(\pi, F^1, F^2) \leq C' \delta_{k,s}$.

Base case ($k = 1$): $\Pi_1 = \Pi$ contains all policies, so $\pi^* \in \Pi_1$ trivially.

Inductive step: Assume the claim holds for epoch k .

Step 1 – Estimation error bound: By Hoeffding’s inequality, over $\tau_k/|A_k|$ rounds per policy, the empirical estimate satisfies:

$$|\bar{F}_{k,e}(i) - F^e(i)| \leq O\left(\sqrt{\frac{\log(4d/\varepsilon)}{M_{k,e}(i)}}\right) \leq O\left(\sqrt{\frac{|A_k| \cdot \log(4d/\varepsilon)}{\tau_k \cdot \pi_{k,i,e}^e(i)}}\right).$$

Since $\tilde{\pi}_{k,i,e}^e(i) \geq 1/\sqrt{T}$ for active prices, and $\tau_k = O(\sqrt{T} \cdot 2^k)$, this gives the $\delta_{k,s}$ bound.

Step 2 – Revenue and unfairness error propagation: From the estimation error on $F^e(i)$, we bound the error in $R(\pi, \hat{F})$ vs. $R(\pi, F)$, and similarly for S . The Lipschitz property of S (guaranteed by Assumption 5.1) is crucial here.

Step 3 – π^ survives elimination:* Since π^* is truly fair ($S(\pi^*, F) = 0$), and our estimate satisfies $|S(\pi^*, \hat{F}) - S(\pi^*, F)| \leq C \delta_{k,s}$, we have $S(\pi^*, \hat{F}) \leq C \delta_{k,s} \leq \delta_{k,s}$ (for appropriate constant C). The revenue check is passed because $\hat{\pi}_{k,*}$ is the best policy in Π_k , so it can’t have revenue exceeding π^* by more than estimation error.

Step 4 – Per-epoch regret: All policies $\pi \in \Pi_{k+1}$ satisfy $R(\pi^*, F) - R(\pi, F) \leq C(\delta_{k,r} + L \delta_{k,s})$, so the per-epoch regret is bounded by $\tau_k \cdot C(\delta_{k,r} + L \delta_{k,s})$.

Step 5 – Summing over epochs:

$$\text{Reg}_T = \sum_{k=1}^K \tau_k \cdot C(\delta_{k,r} + \delta_{k,s}) = \sum_{k=1}^K O\left(\sqrt{T \cdot 2^k} \cdot \sqrt{\frac{1}{\sqrt{T} \cdot 2^k}}\right) = \sum_{k=1}^K O(T^{1/4}) = O(\sqrt{T} \log T).$$

(More careful accounting with the $d^{3/2}$ factors gives the stated bound.)

7.2 Regret Lower Bound

Theorem 7.2 (Regret Lower Bound)

Assume $d \leq T^{1/3}$. For the two-group fair pricing problem, any algorithm must suffer:

$$\text{Reg}_T \geq \Omega(\sqrt{dT}).$$

Proof idea: Reduce to the single-group case. Set $F^1 = F^2$ (identical groups) and $q = 0.5$. Then:

- Any policy with $\pi^1 = \pi^2$ is both procedurally and substantively fair (identical groups $\Rightarrow$ identical accepted prices).
- The optimal strategy is to find the best fixed price.

- This reduces exactly to the standard single-group online pricing bandit problem with a discrete price set.
- Standard bandit lower bounds (Auer et al., 2002) give $\Omega(\sqrt{dT})$ regret for any algorithm.

7.3 Substantive Unfairness Lower Bound

This is the most technically novel lower bound result.

Theorem 7.3 (Substantive Unfairness Lower Bound)

For any constant $C_x > 0$, there exists $C_u > 0$ such that: any algorithm with cumulative regret $\leq C_x \cdot T^{1/2}$ and zero procedural unfairness must satisfy:

$$S_T(\mathcal{A}) \geq C_u \cdot T^{1/2}.$$

Why this is non-trivial: One could trivially achieve zero substantive unfairness by proposing the same price to everyone (which is also procedurally fair). But such a policy would have high regret (since it ignores group-specific demand information). The theorem says: *if you want near-optimal revenue, you must accept $\Omega(\sqrt{T})$ substantive unfairness.*

Proof sketch (the two-environment argument):

Construct two problem instances, $\mathcal{P}$ and $\mathcal{P}'$, both based on Example 4.1 (Chapter 4):

- **Instance $\mathcal{P}$:** Exactly as in Example 4.1. The optimal policy is the random doubly-fair policy with expected accepted price 8/11 in both groups.
- **Instance $\mathcal{P}'$:** Same as $\mathcal{P}$ except the acceptance rates of \$0.7 in G_1 and \$1 in both groups are all changed from 1/2 to $(1/2 - \zeta)$ for a small $\zeta > 0$. This shifts the optimal accepted price slightly.

Four key facts:

1. **No algorithm can do well on both instances simultaneously.** The optimal substantive-fair policies in $\mathcal{P}$ and $\mathcal{P}'$ are different; a policy good for one instance has substantive unfairness on the other.
2. **Instances are information-theoretically indistinguishable for $\Omega(T)$ rounds.** The KL divergence between the two instances is $O(\zeta^2)$ per round. By Pinsker/Fano's inequality, no algorithm can reliably distinguish them in fewer than $O(1/\zeta^2)$ rounds.

3. **Failing to distinguish causes $\Omega(\sqrt{T})$ unfairness.** If the algorithm uses a policy optimal for $\mathcal{P}$ but is actually in $\mathcal{P}'$, the substantive unfairness per round is $\Omega(\zeta)$. Cumulative unfairness over T rounds: $\Omega(\zeta T)$. Setting $\zeta = T^{-1/2}$ gives $\Omega(\sqrt{T})$.
4. **Avoiding the confusion costs even more.** If the algorithm tries to be hedged between the two instances, it cannot achieve the optimal revenue in either one.

Setting $\zeta = C/\sqrt{T}$ for appropriate C , we get $C_u\sqrt{T}$ unfairness as a lower bound for any regret-optimal algorithm.

Chapter 8

Discussion: Extensions and Open Problems

8.1 Technical Assumption Relaxation

Can we remove the $F_{\min} > 0$ assumption?

Two issues arise without it:

1. S might be undefined if some prices are never accepted (zero denominator).
2. The Lipschitz continuity of S depends on $F_{\min}$.

An open problem: can an algorithm handle $F^e(d) = 0$ for some (e, d) while still achieving the same regret and unfairness bounds?

8.2 Relaxed Substantive Fairness

Instead of requiring $S = 0$, consider:

$$\pi_{\delta,*} = \arg \max_{\pi \in \Pi} R(\pi; F^1, F^2) \quad \text{s.t.} \quad S(\pi; F^1, F^2) \leq \delta.$$

Results:

- If $\delta = 0$: regret $\Theta(\sqrt{T})$, unfairness $\Theta(\sqrt{T})$.
- If $\delta \geq 1$: regret $\Theta(\sqrt{T})$, unfairness = 0 (reduces to unconstrained pricing).
- Conjecture for $\delta > \sqrt{1/T}$: regret $\Theta(\sqrt{T})$, unfairness $\Theta(1/(\sqrt{T} \delta))$.
- For $\delta = O(\sqrt{1/T})$: still achievable with both $O(\sqrt{T})$.

8.3 Multi-Group Extension

For G groups, define pairwise unfairness as the sum over all $O(G^2)$ pairs. Extend FPA by lengthening each epoch by $G/2$ times. Resulting upper bound: $\tilde{O}(G^3 \sqrt{T} d^{2/3})$ regret — still optimal in T up to log factors. However, the pairwise multi-group unfairness definition is “somewhat controversial” and alternative metrics exist.

8.4 Continuous Price Space

If prices can be any value in $[0, 1]$ rather than a discrete set V , the problem becomes harder:

- The optimal policy could be a continuous distribution over prices.
- Even knowing F^1, F^2 exactly, computing the optimal continuous doubly-fair policy is an infinite-dimensional optimization problem.
- Discretization methods (Xu & Wang, 2022) might apply but would likely give exponential computational complexity.

8.5 Trade-offs Between Fairness Types

Can we trade procedural unfairness for less substantive unfairness? The authors conjecture: **no significant trade-off exists**. The lower bound on substantive unfairness comes from the fundamental indistinguishability of problem instances, which is not affected by how much procedural fairness we’re willing to sacrifice.

Chapter 9

Implementation Guide

9.1 Setting Up the Problem

To implement FPA, you need to specify:

- **Price set** $V = \{v_1, \dots, v_d\}$: choose these carefully based on domain knowledge.
- **Proportion** q : fraction of customers from G_1 ; can be estimated empirically.
- **Time horizon** T : known in advance or bounded.
- **Error probability** ε : set to, say, 0.05 or 0.01.
- **Constant** L : related to Lipschitz constant of S ; can be bounded as $L \leq 1/F_{\min}^2$.

9.2 Key Implementation Challenges

9.2.1 Linear Programming Subroutine

Both Algorithms 2 and 3 require solving linear programs. Use any standard LP solver (e.g., SciPy's `linprog`, CVXPY, Gurobi, GLPK). The decision variables are $\pi^1, \pi^2 \in \Delta^d$ (probabilities over d prices) and optionally the scalar w .

9.2.2 Grid Search for w

Step through $w_\ell = \ell \cdot \varepsilon_s$ for $\ell = 0, 1, 2, \dots$ where $\varepsilon_s = \delta_{k,s}/2$, up to $w_\ell = 1/\hat{F}_{\min}$. For each w_ℓ , solve the LP (with w fixed) and compare solutions.

9.2.3 Tracking Active Price Sets

Maintain I_k^1 and I_k^2 (sets of prices still actively explored in each group). Update these at the start of each epoch based on whether the maximum policy probability for each price exceeds $1/\sqrt{T}$.

9.3 Complexity Summary

Component	Time Complexity	Space Complexity
Before Epochs	$O(\tau_0) = O(\log T)$	$O(d)$
Good-and-Exploratory	$O(d \cdot \text{LP size})$	$O(d^2)$
Probability Estimation	$O(\tau_k)$ per epoch	$O(d)$
Algorithm 2 (Oracle)	$O(1/\delta_{k,s}) \cdot O(\text{LP})$	$O(d^2)$
Algorithm 3 (Elimination)	$O(1/\delta_{k,s}) \cdot O(\text{LP})$	$O(d^2)$
Total (all epochs)	$O(\text{poly}(T, d, \log T))$	$O(d^2)$

Since $O(\log T)$ epochs are run and each Oracle call solves $O(1/\delta_{k,s}) = \text{poly}(T, d)$ LPs of size $O(d)$, the total time is polynomial in T and d . FPA is **oracle-efficient**.

Chapter 10

Mathematical Appendix: Detailed Proofs

10.1 Derivation of the Optimal Random Policy in Example 4.1

We verify that the stated random policy is indeed doubly fair and compute its optimality (the paper proves it is the *unique* optimal doubly-fair policy in this setting).

Let $\pi^1 = (p, 0, 1 - p)$ over prices $\{0.625, 0.70, 1.00\}$ (Group 1 uses prices 1 and 3 only) and $\pi^2 = (0, s, 1 - s)$ (Group 2 uses prices 2 and 3 only).

Procedural fairness: $0.625p + 1.00(1 - p) = 0.70s + 1.00(1 - s) \Rightarrow 1 - 0.375p = 1 - 0.30s \Rightarrow 0.375p = 0.30s \Rightarrow p = \frac{0.30}{0.375}s = \frac{4}{5}s$.

Substantive fairness:

$$\begin{aligned}\text{Numerator}_1 &= 0.625 \cdot \frac{3}{5} \cdot p + 1.00 \cdot \frac{1}{2} \cdot (1 - p) = 0.375p + 0.5(1 - p) = 0.5 - 0.125p, \\ \text{Denominator}_1 &= \frac{3}{5}p + \frac{1}{2}(1 - p) = 0.5 + 0.1p, \\ \text{Numerator}_2 &= 0.70 \cdot \frac{4}{5} \cdot s + 1.00 \cdot \frac{1}{2} \cdot (1 - s) = 0.56s + 0.5(1 - s) = 0.5 + 0.06s, \\ \text{Denominator}_2 &= \frac{4}{5}s + \frac{1}{2}(1 - s) = 0.5 + 0.3s.\end{aligned}$$

Substantive fairness requires $\frac{0.5 - 0.125p}{0.5 + 0.1p} = \frac{0.5 + 0.06s}{0.5 + 0.3s}$.

Substituting $p = \frac{4s}{5}$:

$$\frac{0.5 - 0.1s}{0.5 + 0.08s} = \frac{0.5 + 0.06s}{0.5 + 0.3s}.$$

Cross-multiplying: $(0.5 - 0.1s)(0.5 + 0.3s) = (0.5 + 0.06s)(0.5 + 0.08s) \Rightarrow 0.25 + 0.15s - 0.05s - 0.03s^2 = 0.25 + 0.04s + 0.03s + 0.0048s^2 \Rightarrow 0.25 + 0.10s - 0.03s^2 = 0.25 + 0.07s + 0.0048s^2 \Rightarrow 0.03s = 0.0348s^2 \Rightarrow s = \frac{0.03}{0.0348} = \frac{25}{29}$ (for $s \neq 0$).

Thus $s = 25/29$ and $p = 4/5 \cdot 25/29 = 20/29$. This exactly matches the policy stated in

the paper!

10.2 Hoeffding's Inequality (Used in Estimation Bounds)

Hoeffding's Inequality

Let $X_1, \dots, X_n$ be independent random variables with $X_i \in [a_i, b_i]$. Then for any $t > 0$:

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] \right| \geq t \right] \leq 2 \exp \left(- \frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

For i.i.d. Bernoulli(p) variables: $\mathbb{P}[|\hat{p} - p| \geq t] \leq 2e^{-2nt^2}$.

Application in FPA: Each observation of “price v_i proposed and accepted/rejected in group G_e ” is a Bernoulli($F^e(i)$) trial. After $M_{k,e}(i)$ observations:

$$\mathbb{P}[|\bar{F}_{k,e}(i) - F^e(i)| \geq t] \leq 2e^{-2M_{k,e}(i)t^2}.$$

Setting $t = \delta_{k,s}/2$ and $M_{k,e}(i) = \Omega(\log(1/\varepsilon)/\delta_{k,s}^2)$ gives the desired high-probability bound.

10.3 Lipschitz Property of Substantive Unfairness

Lemma 9.1 (Lipschitzness of S)

Under Assumption 5.1 ($F_{\min} > 0$), the function $S(\pi; F^1, F^2)$ is Lipschitz in $F^e(i)$:

$$|S(\pi; F) - S(\pi; F')| \leq \frac{2}{F_{\min}^2} \cdot \max_{e,i} |F^e(i) - F'^e(i)|.$$

Proof sketch: The accepted price for group e is $g_e(\pi, F^e) = \frac{\mathbf{v}^\top \mathbf{F}_e \pi^e}{\mathbf{1}^\top \mathbf{F}_e \pi^e}$. The numerator and denominator are both linear in $F^e(i)$. The denominator is bounded below by $F_{\min}$ (by Assumption 5.1). By the quotient rule for Lipschitz functions: if $|h(x)| \leq 1$ and $|g(x)| \geq F_{\min}$, then $|h/g|$ is Lipschitz with constant $\leq (L_h + L_g)/F_{\min}^2$ where L_h, L_g are the Lipschitz constants of numerator and denominator. Both are bounded by d (the number of prices). So $L = O(d/F_{\min}^2)$.

10.4 Why the Policy Set Representation is Tractable

The set Π_k is represented not as an explicit list of policies (impossible, as it's infinite-dimensional) but as a *linear feasibility set* — a set of linear constraints on $(\pi^1, \pi^2) \in \Delta^d \times \Delta^d$.

After Algorithm 3 runs for epoch k , Π_{k+1} is the union of finitely many sets $\Pi_{k+1,w}$, each defined by linear constraints (procedural fairness + revenue bound for fixed w). This union of polytopes can be represented compactly and queried efficiently for the optimization steps in the next epoch.

Chapter 11

Summary: The Complete Picture

11.1 What We Proved

Setup: Two-group online pricing, T rounds, d discrete prices, unknown demand distributions D^1, D^2 .

Key structural insight: Doubly-fair policies must be random (not deterministic) to improve over the trivial equal-price strategy.

Algorithm: FPA (Fairly Pricing Algorithm) — policy elimination with doubling epochs, good-and-exploratory policies, LP-based oracle.

Upper bounds (with probability $\geq 1 - \varepsilon$):

$$\text{Regret} = \tilde{O}\left(\sqrt{T} d^{3/2}\right)$$

$$\text{Procedural Unfairness} = 0 \text{ (exactly, every round)}$$

$$\text{Substantive Unfairness} = \tilde{O}\left(\sqrt{T} d^{3/2}\right)$$

Lower bounds (information-theoretic, any algorithm):

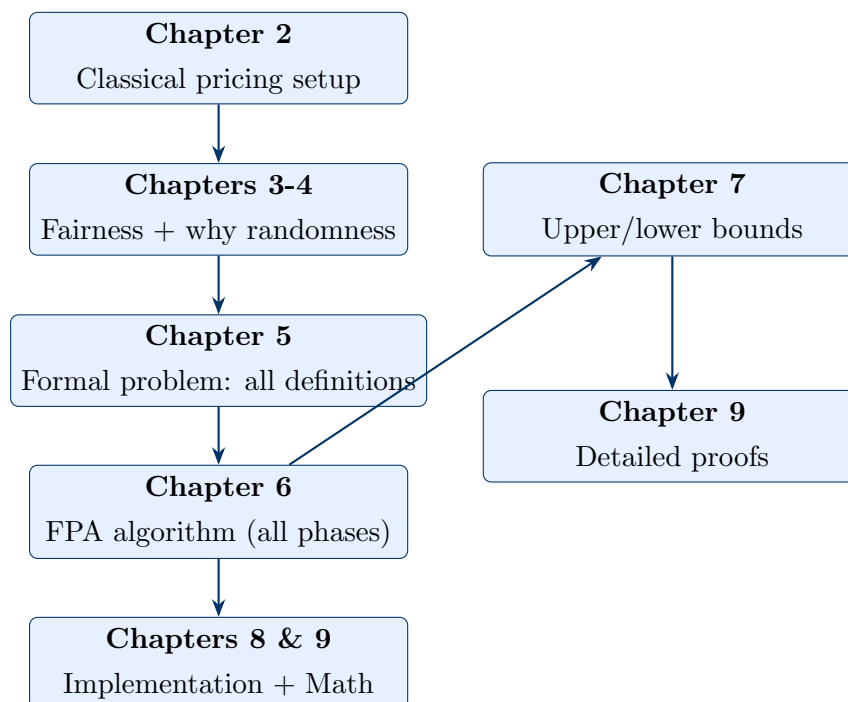
$$\text{Regret} \geq \Omega\left(\sqrt{dT}\right)$$

$$\text{Substantive Unfairness} \geq \Omega\left(\sqrt{T}\right) \text{ (for any regret-optimal algorithm)}$$

Gap: FPA is optimal up to $\tilde{O}(d)$ and $\log \log T$ factors.

Computational cost: Polynomial in T and d (oracle-efficient, $O(\log T)$ LP solves).

11.2 Reading Roadmap for Implementers



11.3 Connections to Broader ML Themes

- **Fairness-accuracy trade-offs:** This paper shows that fairness (substantive) and accuracy (low regret) trade off at rate $\sqrt{T}$, which is the same rate as the regret itself. Fairness does not fundamentally harm learning — it just shifts the constant.
- **Randomization in fair ML:** Just as in classification, deterministic fair policies can be suboptimal, and randomization helps. This is a theme across fair ML theory.
- **Non-convex constrained optimization:** The paper demonstrates a powerful technique — reduce a non-convex problem to a parametric family of LPs via an auxiliary variable. This technique is broadly applicable.
- **Online learning with constraints:** The “conservative elimination” approach (keep π^* in the candidate set at all times, eliminate others as evidence accumulates) is a key technique for constrained online learning.

We want to sell a product to two groups of customers while being fair to both groups in two ways: offering similar prices and resulting in similar actual transaction prices. With fixed prices, the only fair approach is identical prices for everyone — boring and suboptimal. But if we offer *random* price distributions (like a lottery of discounts), we can tune the mix so that both groups see the same *average* offer price and end up paying the same *average* transaction price, while still earning more revenue than any uniform price. The catch is we don't know demand upfront. So we build a learning algorithm that, epoch by epoch, narrows down the set of plausible good-and-fair pricing strategies, using smart exploration to estimate demand and careful “conservative” elimination to avoid throwing away the truly optimal strategy. The math shows this achieves near-optimal revenue with near-zero unfairness, and that neither bound can be substantially improved.